

Portfolio Three: Functions

Program: Animal Island

The purpose of this was to test how games are actually made. I can test here the theories safely and interactive. All concepts were connected from previous and current lesson.

```
In [4]: #starting_animal_island

# Using fruithful function(Returns a calculated relationship)
def calculated_bond(level, bonus):
    if level >= 5:
        return level + bonus
    else:
        return level

# A void function was used to print a news
def show_island_news(text):
    print("ISLAND NEWS: " + str(text))

print("--- Welcome to Animalia! ----")

# Get to know islander profiles
islander1_name = "Missy Cat"
islander2_name = "Rolf Dog"
friend_level = int("3") # String converted to Integer
happiness_percentage = float(80.0) # Integer covertred to float

# Get user input for an action
print("What should they do today?")
print("Action: 'Gift' to give them a present, or 'prank' to cause trouble.")
action = input("Enter your choice: ")

# Using two-way decision for the action
if action == "gift":
    print("You gave them a toy! They played together all day.")
    friend_level = friend_level + 3
else:
    print("Oh no! A thief was spotted! Someone stole the other's favorite snack!")
    friend_level = friend_level - 2

# Calling fruitful function with arguments
final_bond = calculated_bond(friend_level, 2)

# Checking the outcome
if final_bond >= 5:
    happiness_percentage = happiness_percentage + 15.5
    show_island_news(islander1_name + " and " + islander2_name + " became Best friends! Nice one!")
else:
    happiness_percentage = happiness_percentage - 20.0
    show_island_news("Uh ho ... Someone needs an apology.")

# Final checking
print("Island Happiness level: " + str(happiness_percentage))

--- Welcome to Animalia! ----
What should they do today?
Action: 'Gift' to give them a present, or 'prank' to cause trouble.
You gave them a toy! They played together all day.
ISLAND NEWS: Missy Cat and Rolf Dog became Best friends! Nice one!
Island Happiness level: 95.5

In [21]: ## Concepts that were used here:
# Built-in functions
# Type conversion (int,float)
# String conversions
# Parameters
```

```
# Results
# Void functions
# User input
# if/else
```